



ECHO CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & INSTANCE

Installation

```
go get github.com/labstack/echo/v4
e := echo.New()
e.Use(middleware.Logger())
e.Use(middleware.Recover())
e.Start(":8080")
e.Logger.SetLevel(log.DEBUG)
```

04. REQUEST & RESPONSE

Workflow

```
c.Param("id")           // route param
c.QueryParam("page")    // query string
c.FormValue("name")     // form field
return c.JSON(200, user)
return c.String(200, "Hello")
return c.XML(200, xmlData)
return c.File("index.html")
return c.Redirect(301, "/new")
```

02. ROUTING

Architecture

```
e.GET("/users", listUsers)
e.POST("/users", createUser)
e.PUT("/users/:id", updateUser)
e.DELETE("/users/:id", deleteUser)
e.Static("/static", "public")
e.File("/favicon.ico", "public/favicon.ico")
```

05. BINDING & VALIDATION

CRUD API

```
type CreateUserReq struct {
    Name string `json:"name" validate:"required"`
    Email string `json:"email" validate:"required"`
}
var req CreateUserReq
if err := c.Bind(&req); err != nil { return e...
if err := c.Validate(&req); err != nil { retu...
e.Validator = &CustomValidator{v: validator.N...
```

03. MIDDLEWARE

YAML / Config

```
e.Use(middleware.Logger())
e.Use(middleware.Recover())
e.Use(middleware.CORS())
e.Use(middleware.RateLimiter(...))
// Custom:
e.Use(func(next echo.HandlerFunc) echo.Handler...
return func(c echo.Context) error {
    // before
    return next(c)
}
})
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. GROUPING

Hardening

```
g := e.Group("/api/v1")
g.Use(JWTMiddleware())
g.GET("/users", listUsers) // /api/v1/users
admin := g.Group("/admin")
admin.Use(AdminOnly())
admin.GET("/stats", getStats) // /api/v1/admi...
```

10. COMMON CONTEXT METHODS

Unit Tests

```
c.Set("user", user) // store in context
c.Get("user")       // retrieve value
c.Request()         // *http.Request
c.Response()        // *echo.Response
c.RealIP()          // client IP respecting ...
c.Logger()          // scoped logger
c.IsWebSocket()     // check WebSocket upgra...
```

08. STATIC FILES

Observability

```
e.Static("/static", "assets")
e.StaticFS("/static", echo.MustSubFS(os.DirFS...
// Serve index.html for SPA:
e.GET("/*", func(c echo.Context) error {
return c.File("public/index.html")
})
```

11. COMMON GOTCHAS

Commands

Return errors from handlers don't just log and continue
Validator must be set on echo instance before validation
echo.HTTPError message is returned to client don't expose internals
Middleware order matters Logger before Recover
Use c.Set/c.Get for request-scoped data, not global vars

09. ERROR HANDLING

Optimization

```
e.HTTPErrorHandler = func(err error, c echo.C...
code := http.StatusInternalServerError
if he, ok := err.(*echo.HTTPError); ok { code...
c.JSON(code, map[string]string{"error": err.E...
}
return echo.NewHTTPError(http.StatusBadReques...
```